

## CS 217: Artificial Intelligence and Machine Learning

February 26, 2026.

8–10am

Roll: \_\_\_\_\_

Name: \_\_\_\_\_

Write all your answers in the space provided. Do not spend time/space giving irrelevant details or details not asked for. *But you need to show the important steps that are required to justify the answer.* Use the marks as a guideline for the amount of time you should spend on a question. You are only allowed to refer your hand-written notes, no one else's notes or textbook.

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1. An insurance company is trying to classify customers as “Risky” ( $y = 1$ ) or “Not-risky” ( $y = 2$ ) based on two attributes of a customer:  $x_1$  denoting the number of “accidents” in the past ten years and  $x_2$  denoting the type of the vehicle. The table below denotes a sample training set.

|                | $x_1$ | $x_2$ | $y$ |
|----------------|-------|-------|-----|
| $\mathbf{x}^0$ | 0     | F     | 2   |
| $\mathbf{x}^1$ | 3     | T     | 1   |
| $\mathbf{x}^2$ | 1     | T     | 2   |
| $\mathbf{x}^3$ | 2     | S     | 1   |
| $\mathbf{x}^4$ | 2     | F     | 2   |
| $\mathbf{x}^5$ | 3     | T     | 1   |
| $\mathbf{x}^6$ | 4     | S     | 1   |
| $\mathbf{x}^7$ | 5     | S     | 1   |

- (a) Assume for each class  $y$  that the first attribute follows a poisson distribution with parameter  $\lambda_y$  (Recall that for a Poisson distribution  $P(x = k|\lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$ ) Write the maximum likelihood estimate of  $\lambda_y$  for each  $y$ ? [Show the derivation for one  $y$ ] ..2  
The loglikelihood of  $x_1$  in the data given  $y = 1$  is  $\sum_i \log P(x_1^i | y = 1) = \sum_i x_1^i \log \lambda_1 - \lambda_1 = (3 + 2 + 3 + 4 + 5) \log \lambda_1 - 5\lambda_1$  Maximizing this wrt  $\lambda_1$  gives  $\lambda_1 = (3 + 2 + 3 + 4 + 5)/5$ . Similar calculation gives  $\lambda_2 = (0 + 1 + 2)/3$
- (b) Assume the second attribute follows a multinomial distribution with parameter  $\theta_{yr}$  for  $r \in (F, T, S)$ . Write the Bayesian MAP estimate of all the  $\theta_{yr}$ s for  $r \in (F, T, S)$  and each  $y$  assuming the prior distribution is Dirichlet  $\text{Dir}(2, 2, 1)$  on the parameters. ..2  
The map estimation under Dirichlet prior is  $\frac{n_j + \alpha_j}{\sum_{i=1}^K (n_i + \alpha_i)}$ . Applying this formula to the data of each class gives  
 $\theta_{1F} = (0 + 2)/(5 + 5)$ ,  $\theta_{2F} = (2 + 2)/(3 + 5)$   
 $\theta_{1T} = (2 + 2)/(5 + 5)$ ,  $\theta_{2T} = (1 + 2)/(3 + 5)$   
 $\theta_{1S} = (3 + 1)/(5 + 5)$ ,  $\theta_{2S} = (0 + 1)/(3 + 5)$
- (c) Now say we try to create a decision tree. If the first split is on  $x_2$  draw the three branches and the number of examples in each class after the split. ..1 **Easy**
- (d) What is the information gain of the above split? [No need to use a calculator to give a single number, you can leave the answer in terms of logs.] ..1 **Easy**

- (e) If the first split is on  $x_1 \leq 2$  draw the two branches and the number of examples in each class after the split. ..1 Easy
- (f) What is the information gain of the above split? ..1 Easy

2. Consider this  $\epsilon$  insensitive regression loss formulation where given  $\hat{y} = f(\mathbf{x}) = \mathbf{w}\mathbf{x} + w_0$ , the training loss is defined as

$$L_\epsilon(y, \hat{y}) \triangleq \begin{cases} 0 & \text{if } |y - \hat{y}| < \epsilon \\ |y - \hat{y}| - \epsilon & \text{otherwise} \end{cases}$$

(a) Show that the objective below

$$\min_{\mathbf{w}, w_0} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{n=1}^N L_\epsilon(y_n, f(\mathbf{x}_n)) \quad (1)$$

is equivalent to:

$$\min_{\mathbf{w}, w_0, \xi_n^+, \xi_n^-} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{n=1}^N (\xi_n^+ + \xi_n^-)$$

such that

$$\begin{aligned} y_n &\leq f(x_n) + \epsilon + \xi_n^+ \\ y_n &\geq f(x_n) - \epsilon - \xi_n^- \\ \xi_n^+ &\geq 0, \quad \xi_n^- \geq 0 \end{aligned} \quad (2)$$

..2 Using arguments similar to what we developed for SVM it is easy to show that  $\xi_n^+ = \max(0, y_n - f(x_n) - \epsilon)$  and  $\xi_n^- = \max(0, -y_n + f(x_n) - \epsilon)$  because of the constraints and the minimization objective.

if  $y > \hat{y} + \epsilon$  then loss is  $y - \hat{y} - \epsilon$  and we get that  $\xi_n^+ = y_n - f(x_n) - \epsilon$  and  $\xi_n^- = 0$

(b) Consider an example, for which  $y_n = 1, f(x_n) = 0, \epsilon = 0.5$ . Write down the value of  $\xi_n^+$  and  $\xi_n^-$  ..1 Everyone should get this.

(c) The dual of the above objective can be written as

$$\begin{aligned} \max_{\alpha_n^+, \alpha_n^-} \quad & \sum_{n=1}^N y_n (\alpha_n^+ - \alpha_n^-) - \epsilon \sum_{n=1}^N (\alpha_n^+ + \alpha_n^-) \\ & - \frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N (\alpha_n^+ - \alpha_n^-) (\alpha_m^+ - \alpha_m^-) \mathbf{x}_n^T \mathbf{x}_m \\ \text{subject to} \quad & \sum_{n=1}^N (\alpha_n^+ - \alpha_n^-) = 0 \\ & \alpha_n^+ + \mu_n^+ = C, \quad \alpha_n^- + \mu_n^- = C, \quad \alpha_n^+, \mu_n^+, \alpha_n^-, \mu_n^- \geq 0 \end{aligned} \quad (3)$$

In the above the mapping between the dual variables and the primal constraints follows notation similar to what we discussed for SVMs for classification. Using this dual write the relevant KKT conditions and for the following three cases reason about the values of  $\alpha, \xi$  variables. Marks will be awarded only if the reasoning is correct.

i. Case 1:  $y_n < f(x_n) - \epsilon$ .

..2 The KKT conditions relevant here are:

$$\alpha_n^+(\epsilon + \xi_n^+ + f(x_n) - y_n) = 0 \quad (4)$$

$$\alpha_n^-(\epsilon + \xi_n^- - f(x_n) + y_n) = 0 \quad (5)$$

$$\mu_n^+ \xi_n^+ = 0 \quad (6)$$

$$\mu_n^- \xi_n^- = 0 \quad (7)$$

$$\mu_n^+ + \xi_n^+ = C \quad (8)$$

$$\mu_n^- + \xi_n^- = C \quad (9)$$

$$(10)$$

The loss  $\xi_n^- > 0$ . This implies that  $\mu_n^- = 0$ , and  $\alpha_n^- = C$ . Also,  $\xi_n^+ = 0$  and from KKT 4 we get  $\alpha_n^+ = 0$ .

ii. Case 2:  $|y_n - f(x_n)| < \epsilon$ .

..2 From first two kkt conditions we get that

$$\alpha_n^- = \alpha_n^+ = 0$$

iii. Case 3:  $y_n = f(x_n) + \epsilon$ .

..2 Here we know that

$\xi_n^- = 0, \xi_n^+ = 0$  and  $(\epsilon + \xi_n^+ + f(x_n) - y_n) = 0$  but  $(\epsilon + \xi_n^- - f(x_n) + y_n) \neq 0$ . Thus from second kkt condition  $\alpha_n^- = 0$ . Nothing else is known.

3. Consider a new kind of classifier training dataset where for each instance  $\mathbf{x}^i$  we are given a parameter  $o_i$  such that the true probability over labels  $y$  of  $\mathbf{x}^i$  is  $\tilde{P}(y = 1|\mathbf{x}^i) = \frac{1}{1+e^{-o_i}}$  and  $\tilde{P}(y = -1|\mathbf{x}^i) = 1 - \tilde{P}(y = 1|\mathbf{x}^i)$ . An example, of such training data for  $d = 2$  is  $D_3 = \{([1, 2]', -1), ([0, 1]', -2), ([2, 1]', 0)\}$ . Suppose we now train a classifier  $C(\mathbf{x}) = \text{sign}(\mathbf{w}^T \mathbf{x} + w_0)$  to fit this dataset well.

(a) Propose a good set of values for  $\mathbf{w} = [w_1 \ w_2]'$ ,  $w_0$  on the example dataset  $D_3$  assuming no regularizer. ..3

We attempt to fit the values of  $w_1, w_2, w_0$  so that  $\mathbf{w}^T \mathbf{x}^i + w_0 = o_i$  for  $i = 1, 2, 3$ . This can be easily found to be  $[w_1 \ w_2]' = [1 \ -2]'$ ,  $w_0 = 0$

(b) Suppose we train the dataset using the following objective

$$\min_{\mathbf{w}, w_0} \sum_{i=1}^N \text{klDist}(\tilde{P}(y|\mathbf{x}^i), \hat{P}(y|\mathbf{x}^i, \mathbf{w}, w_0))$$

where  $\hat{P}(y = 1|\mathbf{x}^i, \mathbf{w}, w_0) = \frac{1}{1+e^{-\mathbf{w}^T \mathbf{x}^i - w_0}}$  and  $\text{klDist}(p(y), q(y))$  is a function that measures the KL distance between two distributions as  $\sum_{y \in \{-1, 1\}} p(y) \log \frac{p(y)}{q(y)}$ . Show that this training objective is convex in  $\mathbf{w}$  and  $w_0$ . You can use known results discussed in class for this proof. ..3

Easy, the objective is equivalent to a logistic loss objective with the only difference being we have a positive weighted combination of two terms.

4. Suppose you apply a LDA classifier and a naive Bayes classifier (with a Gaussian distribution) on a 1-d dataset where the *true* data distribution is a mixture of two Gaussian distributions. That is,

$$\Pr(x|y) = \alpha_y N(x; \mu = m_1, \sigma^2 = s_1^2) + (1 - \alpha_y) N(x; \mu = m_2, \sigma^2 = s_2^2).$$

where  $0 \leq \alpha_y \leq 1$  Thus the true distributions of the two classes differ in only the proportion ( $\alpha_y$ ) in which the two Gaussians are mixed. Assume that you have infinite samples from the

true distribution as the training dataset  $D$ . Let  $\pi_1$  denote the fraction of class 1 instances during training.

- (a) You use  $D$  to train a naive Bayes model where we fit  $\hat{P}(x|y) = N(x; \mu_y, \sigma_y^2)$ . Note, here during training we fit a single Gaussian for each class although the true data distribution is a mixture. Using  $D$  what will be the maximum likelihood estimate of parameters  $\mu_1, \mu_2, \sigma_1, \sigma_2$  in terms of given values of  $m_1, m_2, s_1, s_2, \alpha_1, \alpha_2$ . assuming  $\pi_1 = 1/3$ .

i.  $\mu_1$  ..1

$$\alpha_1 m_1 + (1 - \alpha_1) m_2$$

ii.  $\mu_2$  ..1

$$\alpha_2 m_1 + (1 - \alpha_2) m_2$$

iii.  $\sigma_1^2$  ..1

$$\alpha_1 s_1^2 + (1 - \alpha_1) s_2^2 + \alpha_1 (1 - \alpha_1) (m_1 - m_2)^2$$

iv.  $\sigma_2^2$  ..1

$$\alpha_2 s_1^2 + (1 - \alpha_2) s_2^2 + \alpha_2 (1 - \alpha_2) (m_1 - m_2)^2$$

- (b) Next if we fit an LDA model in the standard form that is where  $\hat{P}(x|y) = N(x; \mu_y, \sigma^2)$ . What is  $\sigma^2$ 's MLE estimate? ..2

$$\pi_1 \sigma_1^2 + (1 - \pi_1) \sigma_2^2$$

5. Standard Ordinary Least Squares (OLS) regression is derived from the assumption that the target variable  $y$  is a linear combination of features plus Gaussian noise:  $y = w^T x + \epsilon$ , where  $\epsilon \sim \mathcal{N}(0, \sigma^2)$ .

Suppose you are working on a problem where the noise  $\epsilon$  follows a Uniform distribution  $U(-\lambda, \lambda)$  instead of a Gaussian distribution. Assume  $\lambda > 0$  is a known constant.

- (a) Write down the probability distribution  $P(y^{(i)}|x^{(i)}; w)$ . Hint: Remember the constraint on  $\epsilon$  imposed by the uniform distribution. ..3

$$P(y^{(i)}|x^{(i)}; w) = \begin{cases} \frac{1}{2\lambda} & \text{if } |y^{(i)} - w^T x^{(i)}| \leq \lambda \\ 0 & \text{otherwise} \end{cases} \quad (11)$$

- (b) If you are given a training set  $D = \{(1, 4), (-1, -3.5), (2, 5)\}$  and  $\lambda = 1$ , what is the optimal value of  $w$ ? ..3 For the likelihood to be non-zero, the weight  $w$  must satisfy the constraint  $|y^{(i)} - w x^{(i)}| \leq 1$  for all points  $i = 1, 2, 3$ . This generates a system of inequalities:

i.  $|4 - w(1)| \leq 1 \implies 3 \leq w \leq 5$

ii.  $|-3.5 - w(-1)| \leq 1 \implies |-3.5 + w| \leq 1 \implies 2.5 \leq w \leq 4.5$

iii.  $|5 - w(2)| \leq 1 \implies -1 \leq 5 - 2w \leq 1 \implies 2 \leq w \leq 3$

To find the optimal  $w$ , we look for the intersection of these intervals:

$$w \in [3, 5] \cap [2.5, 4.5] \cap [2, 3] \implies w = 3 \quad (12)$$

The only value that satisfies all constraints simultaneously is  $w = 3$ .

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| <b>Total: 35</b> |
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